Hamiltonian Dynamics and Three-body Problem

Dongho Lee

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SNU, QSMS
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Introduction

The **three-body problem**, came from the celestial mechanics, describes the motion of an object under the gravitational force of two bodies.

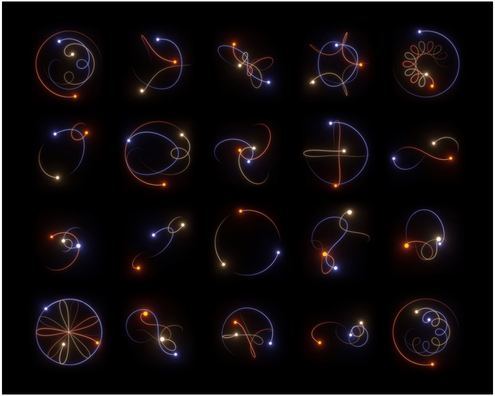

Figure 1: Various solutions of three-body problem.[1]

[1] https://www.wikiwand.com/en/articles/Three-body_problem

Contents

Main interest : Periodic orbits of three-body problem.

- Hamiltonian formalism and symplectic geometry
- Morse theory and Floer theory
- Bifurcation of periodic orbits

Hamiltonian Dynamics and Symplectic Geometry

Classical Mechanics

Classical mechanics is governed by 2nd order ODE on $\mathbb{R}^n$ which is given by Newton's second law,

$$F = ma \quad \Leftrightarrow \quad -\nabla V(q) = \ddot{q},$$

where $q \in \mathbb{R}^n$ is the position and V is the **potential energy**.

The **mechanical energy** is defined by

$$H = K + V := \frac{|\dot{q}|^2}{2} + V(q),$$

where K is the **kinetic energy**.

Law (Classical Energy Consevation)

H is conserved along the trajectory of q determined by force F of the form $-\nabla V$ where $V = V(q)$.

Hamiltonian Formalism

Question 1. Can we re-formulate the equation of motion via the law of energy conservation?

Consider H as a function $H = H(q, p)$ where $p \in T_q^*\mathbb{R}^n$, so

$$H = \frac{|p|^2}{2} + V(q).$$

Then we have

$$\dot{q} = p = \frac{\partial H}{\partial p},$$
$$\dot{p} = \ddot{q} = -\nabla V = -\frac{\partial H}{\partial q}.$$

Now we consider 1st order ODE on $T^*\mathbb{R}^n$.

Hamiltonian Formalism

Let $H : T^*\mathbb{R}^n \to \mathbb{R}$ be a function, which will be called **Hamiltonian**.

Consider the vector field X_H on $T^*\mathbb{R}^n$,

$$X_H = \begin{pmatrix} \partial_p H \\ -\partial_q H \end{pmatrix} = \sum_i \frac{\partial H}{\partial p_i}\frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial}{\partial p_i}.$$

Theorem (Energy Conservation)

Along the trajectory of X_H, H is conserved.

We call X_H the **Hamiltonian vector field**.

Now we can impose *any* function H as the total energy of the system, and use the viewpoint of the classical mechanics.

Symplectic Structure

Question 2. How can we make X_H from H?

We have

$$dH = \sum_i \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i,$$

so if we can change (dq, dp) to $(-\partial_p, \partial_q)$, we get X_H.

The duality via $\omega = dp \wedge dq$ works by

$$\omega(X_H, \cdot) = -dH(\cdot),$$

since

$$\omega(X_H, \partial_{q_i}) = \omega\left((-\partial_{q_i} H)\, \partial_{p_i}, \partial_{q_i}\right) = -\partial_{p_i} H = -dH(\partial_{q_i}),$$
$$\omega(X_H, \partial_{p_i}) = \omega\left((\partial_{p_i} H)\, \partial_{q_i}, \partial_{p_i}\right) = -\partial_{p_i} H = -dH(\partial_{p_i}).$$

Symplectic Structure

For suitable 2-tensor ω, we must have

1. **Non-degeneracy** ($\ker \omega = 0$) for duality.
2. **Alternating property** ($\omega(X, X) = 0$) for conservation of energy,

$$X_H(H) = dH(X_H) = -\omega(X_H, X_H) = 0.$$

3. **Closedness** ($d\omega = 0$) for the conservation of ω,

$$\mathcal{L}_{X_H}\omega = di_{X_H}\omega + i_{X_H}d\omega = d(dH) + i_{X_H}d\omega = i_{X_H}d\omega = 0.$$

We call closed non-degenerate 2-form ω a **symplectic form**.

Symplectic Geometry

A manifold W with a symplectic form ω is called **symplectic manifold**.

The study of (W, ω) is symplectic geometry / topology / dynamics.

Basic Properties

1. $(T^*M, dp \wedge dq)$ is a typical example of symplectic manifold.
2. $\dim W = 2n$, since ω must be non-degenerate.
3. $\omega^n = \omega \wedge \cdots \wedge \omega$ is a volume form, and thus W is orientable.
4. If W is compact, $H^2(M) \neq 0$ since $d\omega = 0$ and ω cannot be exact by Stokes' theorem.

Nice introductory texts : [Arn89], [CdS01].

Bifurcation and Index

Periodic Orbits

Main interest : **Periodic orbits** of a given Hamiltonian,

$\gamma : [0, \tau] \to W$ such that $\dot{\gamma}(t) = X_H(\gamma(t))$, $\gamma(0) = \gamma(\tau)$.

How many are there? How do they behavior?

- Critical point of the **action functional** defined on the loop space.
- If $f : M \to \mathbb{R}$ is smooth Morse function, $\#\mathrm{Crit}(f) \geq \sum \dim H_i(M)$.

  $\Rightarrow$ Arnold Conjecture : $\#\{$Periodic Orbits$\} \geq \sum \dim H_i(W)$

  $\Rightarrow$ Connection between (symplectic) topology and dynamics.
- Generator of a **Floer chain complex**.

  $\Rightarrow$ Building block of symplectic topology.
- Practical purposes. (In next talk)

Toy Example : Function

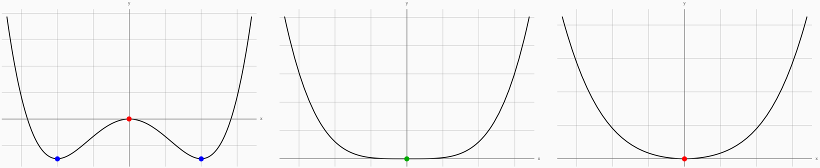

Figure 2: Toy model of a bifurcation and CZ-index

Consider the family of functions $f_t(x) = x^4 - tx^2$.

As t varies, a family of critical points changes when it degenerates.

Conley-Zehnder Index

A periodic orbit γ is **non-degenerate** if there is no *multiplicity*.

(Precise definition : $\det(dFl_\tau^{X_H}|_{H^{-1}(c)}(q) - \mathrm{Id}) = \langle X_H \rangle$.)

Generically, every periodic orbit of X_H on $H^{-1}(c)$ is non-degenerate.

For non-degenerate γ, we define **Conley-Zehnder index** $\mu_{CZ}(\gamma) \in \mathbb{Z}$.

- Can be regarded as a 'symplectic rotation number'.
- Gives grading of **Floer homology**, which is a symplectic invariant.
  $\Rightarrow$ The indices of *all* periodic orbits are topologically controlled.
- Tools: local Floer homology, Morse-Bott spectral sequence, etc.

Bifurcation of Periodic Orbits

Let γ_c be a periodic orbit of X_H in $H^{-1}(c)$.

1. If γ_c is non-degenerate, there is an *orbit cylinder*

$$\Gamma : S^1 \times (c - \varepsilon, c + \varepsilon) \to W$$

   such that $\Gamma(\cdot, c') = \gamma_{c'}(\cdot)$ is a periodic orbit of X_H in $H^{-1}(c')$.

2. If γ_c is non-degenerate for $c_0 \leq c \leq c_1$, the CZ-index of γ_c for $c_0 \leq c \leq c_1$ is constant.
3. If γ_c becomes degenerate, the index might change and new families of orbit might born, or the family γ_c might vanish.
   We call this **bifurcation**.
4. Floer homology is invariant of the choice of c unless $c \in \mathrm{Crit}(H)$, so we can track the bifurcation by computing Conley-Zehnder indices.

Bifurcation of Periodic Orbits

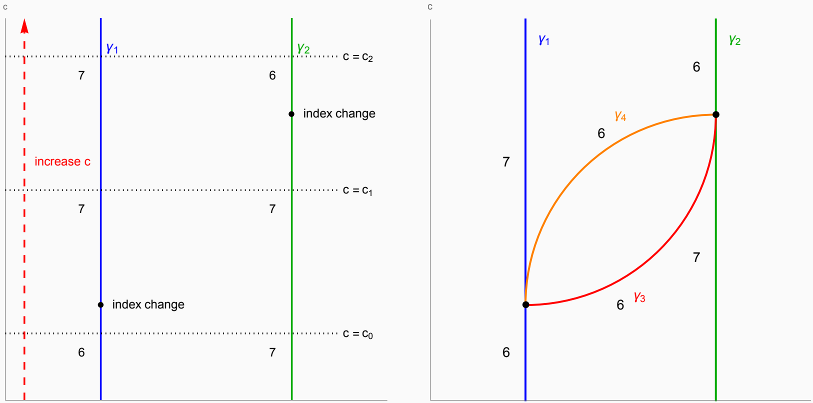

Figure 3: Toy model of a bifurcation and CZ-index

Bifurcation of Periodic Orbits

Remarks

- For degenerate orbits, we can define Robbin-Salamon index $\mu_{RS} \in \frac{1}{2}\mathbb{Z}$.
- Instead of changing the energy, we can perturb the Hamiltonian and see the bifurcation behavior.

  $\Rightarrow$ Investigate a complicated system as a perturbation of simpler one.

Nice introductory texts

- Morse Theory : [Mil63]
- Floer Theory : [AD14], [Abo15], [KvK16]
- Bifurcations : [AM78]

Three-body Problem

Kepler Problem

The **Kepler problem** describes the motion of an object under the gravitational force of the other body.

$$E(q,p) = \frac{|p|^2}{2} - \frac{1}{|q|}.$$

In 17th century, Johannes Kepler established three laws from pure observation...

Theorem (Kepler's Laws)

1. *The E-orbits are conic sections with one focus at the origin. If $E < 0$, the orbits are ellipses.*
2. *The areal velocity $\mathrm{dArea} = r^2\dot{\theta}$ is constant along the orbit.*
3. *The period is given by $\tau = 2\pi/(-2E)^{3/2}$.*

Setting

Let two bodies E (earth) and M (moon) has mass ratio $1-\mu : \mu$.

$$H_t(q,p) = \frac{|p|^2}{2} - \frac{\mu}{|q-M(t)|} - \frac{1-\mu}{|q-E(t)|}$$

Assumptions

1. The third body is much lighter than M and E, so the motion of M and E are not affected by the third body.

   $\Rightarrow$ The motions of M and E are governed by Kepler problem.
2. M and E are in *circular* motion.

Circular Restricted Three-body Problem

The **circular restricted three-body problem** (CRTBP) is determined by the same $H_t(q,p)$ with

$$M(t) = (1-\mu)(\cos t, -\sin t, 0), \quad E(t) = -\mu(\cos t, -\sin t, 0).$$

We can remove the time-dependency by using *rotating frame*, which gives

$$H(q,p) = \frac{|p|^2}{2} - \frac{\mu}{|q-M|} - \frac{1-\mu}{|q-E|} + (p_1 q_2 - p_2 q_1).$$

This time-independent Hamiltonian also defines CRTBP.

Circular Restricted Three-body Problem

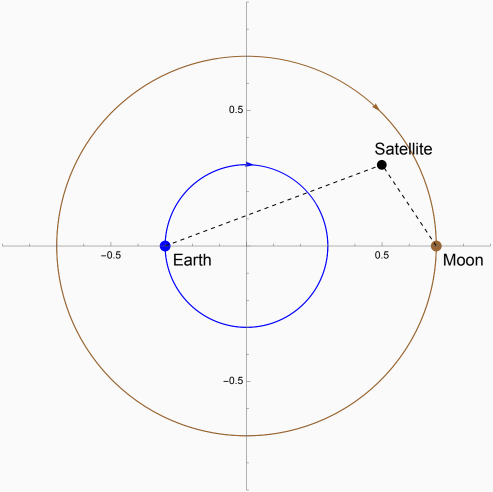

Figure 4: Illustration of CRTBP.

Rotating Kepler Problem

Main interest : Periodic orbits of CRTBP.

CRTBP is still hard... $\Rightarrow$ Use limit cases to get ideas.

Rotating Kepler problem (RKP) is the limit case $\mu = 0$:

$$H(q,p) = \frac{|p|^2}{2} - \frac{1}{|q|} + (p_1 q_2 - p_2 q_1).$$

Note. The trajectory can be decomposed into the Kepler trajectory and the rotation along q_3-axis.

Periodic Orbits of RKP

There are 4 non-degenerate periodic orbits of RKP for generic energy.

- Planar circular orbits : **retrograde orbit** γ_+ and **direct orbit** γ_-.
- **Vertical collision orbits** : north and south $\gamma_{c_\pm}$.

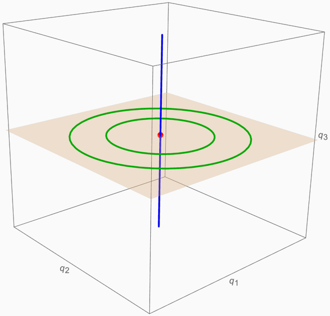

Figure 5: Non-degenerate orbits of RKP.

Bifurcation of RKP

Let $E_{k,l} = -\frac{1}{2}(\frac{k}{l})^{2/3}$ and

$$c^{\pm}_{k,l} = E_{k,l} \pm \frac{1}{\sqrt{-2E_{k,l}}}.$$

We consider the change of energy c.

1. $c = c^{-}_{k,l}$: γ^{k-l}_{-} become degenerate.
2. $c = E_{k,l}$: $\gamma^{k}_{c_{\pm}}$ become degenerate, but no bifurcation.
3. $c = c^{+}_{k,l}$: γ^{k+l}_{+} become degenerate.
4. An S^3-family of orbits $\Sigma_{k,l}$ borns at $c^{-}_{k,l}$ and vanishes at $c^{+}_{k,l}$.

Bifurcation of RKP

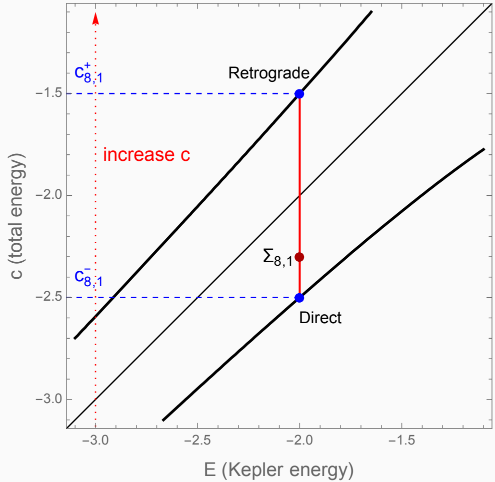

Figure 6: Bifurcation diagram.

CZ Index of RKP

Theorem ([Lee25], ArXiv preprint)

Orbits	Initial Index	Index Change
Retrograde γ_+^N	$\mu_{CZ} = 4N - 2$ *if* $c < c^+_{N-1,1}$	-4 *at* $c = c^+_{N-k,k}$ *for* $k = 1, \ldots, N-1.$
Direct γ_-^N	$\mu_{CZ} = 4N + 2$ *if* $c < c^-_{N+1,1}$	$+4$ *at* $c = c^-_{N+k,k}$ *for* $k = 1, 2, \ldots$
Vertical Collisions $\gamma_{c_\pm}^N$	$\mu_{CZ} = 4N$	*No change*
$\Sigma_{k,l}$*-family*	$\mu_{RS} = 4k - 1/2$	-

[AFFvK13] computed the index for the planar problem, and the result for $\gamma_\pm^N$ is exactly the half.

Bifurcation of RKP

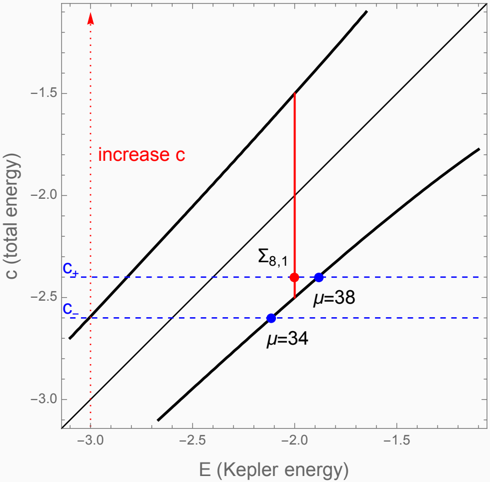

Figure 7: Bifurcation and the change of indices.

Final Remarks

- We can consider CRTBP as a perturbation of RKP, and consider the family of non-degenerate orbits.
- 4 non-degenerate orbits of RKP also exist in CRTBP, which can be shown numerically.
- CRTBP has much less symmetry, so $\Sigma_{k,l}$-family does not exist, but new periodic orbits might exist.

Nice introductory text : [Cel10], [FvK18]

Thank you for your attention!

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